

Oscar Pueyo-Ciudad

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Resume

I am Oscar Pueyo-Ciudad, PhD Student in Computational Imaging supervised by [Prof. Diego Gutierrez](#) and [Prof. Albert Redo-Sanchez](#) at the [Graphics and Imaging Lab](#), Universidad de Zaragoza (Spain). Previously, I completed a Bachelor of Computer Engineering and a Master of Engineering in Robotics, Graphics, and Computer Vision from the Universidad de Zaragoza. During my PhD, I have visited [Princeton Computational Imaging Lab](#) for three months, supervised by [Prof. Felix Heide](#).

My research interests are related to Computational Imaging: Transient and polarization rendering/imaging, wave optics and holography, and forward and inverse physically-based rendering.

Publications

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| Statistical Denoising of Transient Rendering | 2026 |
| Oscar Pueyo-Ciudad, Alvaro Lopez, Diego Gutierrez
10.1111/cgf.70321 (Computer Graphics Forum (EUROGRAPHICS 2026)) | |
| Artifact-Resilient Real-Time Holography | 2025 |
| Victor Chu, Oscar Pueyo-Ciudad, Ethan Tseng, Florian Schiffers, Grace Kuo, Nathan Matsuda, Albert Redo-Sanchez, Douglas Lanman, Oliver Cossairt, Felix Heide
10.1145/3763361 (SIGGRAPH Asia 2025 (Journal track)) | |
| Time-Gated Polarization for Active Non-Line-Of-Sight Imaging | 2024 |
| Oscar Pueyo-Ciudad, Julio Marco, Stephane Schertzer, Frank Christnacher, Martin Laurenzis, Diego Gutierrez, Albert Redo-Sanchez
10.1145/3680528.3687575 (SIGGRAPH Asia 2024) | |

Education

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|--|-----------------------|
| Ph.D. Universidad de Zaragoza , Computational Imaging | Jan 2024 – present |
| • Supervised by Prof. Diego Gutierrez and Prof. Albert Redo-Sanchez. | |
| MEng Universidad de Zaragoza , Robotics, Graphics and Computer Vision | Sept 2022 – Feb 2024 |
| • GPA: 9.73 / 10.0. #1 of the promotion.
• Specialization in Computational Imaging. | |
| BEng Universidad de Zaragoza , Computer Engineering | Sept 2018 – July 2022 |
| • GPA: 9.63 / 10.0. #1 of the promotion.
• Specialization in Computing. | |

Projects

- mitransient. Transient light transport in Mitsuba 3**
- Collaborated in the [mitransient](#) project, added support for time-resolved polarization rendering ([docs](#)).
- Exploiting polarization in Non-Line-of-Sight Imaging**
- Master's thesis. Research on adding and exploiting the information encoded in polarization for NLOS imaging.
 - Supervised by Prof. Albert Redo-Sanchez and Prof. Diego Gutierrez.
 - Grade: 10.0 / 10.0 with honors.
- Place recognition in Visual SLAM with endoscopic sequences**

- Bachelor's thesis. Using computer vision point features and bags of words in appearance based topological SLAM in endoscopic sequences ([EndoMapper project](#)).
- Supervised by [Juan D. Tardos](#) and [Juan J. Gomez](#).
- Grade: 9.8 / 10.0 with honors.

Experience

Princeton Computational Imaging Lab , Visiting Researcher	Princeton (NJ), USA
• Three-month visit under the supervision of Prof. Felix Heide.	Mar 2025 – May 2025
POSTECH Computational Imaging Lab , Visiting Researcher	Pohang, South Korea
• One-week visit for an ongoing collaboration.	Dec 2024
French-German Research Institute of Saint-Louis (ISL) , Visiting Researcher	Saint Louis, France
• One-week visit for NLOS experimental captures with time-resolved systems.	Dec 2024
Graphics and Imaging Lab , Research engineer	Zaragoza, Spain
• Worked on integrating polarization into NLOS imaging.	Sept 2023 – Jan 2024
Graphics and Imaging Lab , Research intern	Zaragoza, Spain
• Studied transient rendering and simulation of hidden scenes, polarization of light, and the basics of non-line-of-sight imaging.	Mar 2023 – June 2023
• Developed prototype algorithms combining NLOS with polarization.	
Robotics, Perception and Real Time (RoPeRT) , Scholarship researcher	Zaragoza, Spain
• Studied fundamentals of Computer Vision and developed a topological Visual SLAM algorithm to guide the surgeon during endoscopies.	Sept 2021 – July 2022

Skills and Technologies

Wave-optics and holography: Deep learning and gradient-based optimization for Computer-Generated Holography, and fundamentals of light-wave propagation. [PyTorch](#), [PADO](#).

Non-line-of-sight (NLOS) imaging: Reconstruction and simulation of scenes hidden around a corner, developed a reconstruction algorithm integrating time-gated polarization to NLOS. [Mitsuba3](#) & [Dr.JiT](#), [mitransient](#), [TAL](#).

Transient rendering & imaging: Simulation and capture of transient data. [Mitsuba3](#) & [Dr.JiT](#), [mitransient](#).

Polarization rendering & imaging: Reflectance models, Monte Carlo integration, differentiable rendering, ellipsometry and Mueller/Stokes calculus. [Mitsuba3](#) & [Dr.JiT](#).

Computer vision / Simultaneous Localization and Mapping: Traditional and learned feature matching, Structure from Motion, Bag of Words representation, SLAM. [OpenCV](#), [DBow](#), [ORB-SLAM3](#).

Awards and Acknowledgements

AGM Award to the best academic record	June 2025
• Award for the best academic results of all the Master's Degrees in Engineering at the Universidad de Zaragoza.	
Extraordinary End-of-Master Award	June 2025
• Award for the best academic results during the Master's Degree in Robotics, Graphics and Computer Vision.	
FPU Grant	Jan 2024 – Jan 2028
• Most competitive Spanish grant for funding the Ph.D.	
• Four-year grant from the Spanish Ministerio de Ciencia, Innovación y Universidades.	

Extraordinary End-of-Degree Award	June 2023
Award for the best academic results during the Computer Engineering Degree.	
Best Bachelor's thesis of Computer Engineering in Aragon	Apr 2023
First prize in the Colegio Profesional de Ingeniería Informática de Aragón best Bachelor's thesis awards for my Bachelor's thesis Place Recognition in Visual SLAM with endoscopic sequences, where we explored Computer Vision and SLAM techniques to build a topological map to guide the surgeon in colonoscopies.	
Honors in 5 subjects during the Master of Engineering	Sept 2022 – Feb 2024
Honors in 5 subjects during the Master of Engineering in Robotics, Graphics, and Computer Vision, including the Master's thesis.	
Honors in 32 subjects during the Bachelor of Computer Engineering	Sept 2018 – July 2022
Honors in 32 out of 39 subjects during the Bachelor of Computer Engineering, including the Bachelor's thesis.	

Dissemination

Thesis in 3 minutes	Oct 2025
I presented my research to the general public in only 3 minutes.	
Researcher's night	Sept 2024
We explained our research at the Graphics and Imaging Lab to the general public.	
Week of Engineering and Architecture	Mar 2024
We showed experiments to high-school students on light transport and graphics.	
Researcher's night	Sept 2023
We explained our research at the Graphics and Imaging Lab to the general public.	

Languages

English (CAE - C1)	2018
Cambridge English Level 2 Certificate in ESOL International (Advanced). Grade 186	
Spanish (native)	

Social Skills

Class delegate	2018 – 2022
Delegate of our class during the Bachelor Degree of Computer Engineering.	
Degree delegate	2019 – 2021
Delegate of the Bachelor Degree in Computer Engineering.	

Sports

Volleyball player - AD Cierzo proLGTB+	2025 – present
Volleyball player - AD Miguel Catalán	2023 – 2025
Volleyball player - Club Voleibol Zaragoza	2019 – 2023
Volleyball player - School of engineering and architecture	2019 – 2023